Chapter 3. Common Distributions

Hypergeometric, binomial, multinomial, geometric, negative binomial, Poisson, exponential, gamma, normal

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Conventions

Many distributions below are introduced through a **representation**: $X \overset{d}{=} Z_1 + \cdots + Z_n$ means X has the same distribution as a sum of the indicated variables. A representation makes means, variances and mgfs almost free (Theorem 2.10 and the product rule for mgfs of independent sums). Throughout, $\mathrm{Exp}(\beta)$ and $\mathrm{Gamma}(\alpha, \beta)$ use the *scale* β, so their means are β and $\alpha\beta$.

3.1 The hypergeometric distribution

A population of N units contains D of type "1" and $N-D$ of type "0". The population proportion is $p = D/N$. Draw n units by **simple random sampling**: one at a time, each remaining unit equally likely, without replacement. The number X of type-1 units in the sample has the **hypergeometric distribution** $X \sim H(n; N, D)$:

$$f(x) = \binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n}, \qquad 0 \le x \le D,\ 0 \le n-x \le N-D.$$

Proposition 3.1

$EX = np$ and $\operatorname{Var} X = \dfrac{N-n}{N-1}\,np(1-p)$.

Proof. Using $x\binom{D}{x} = D\binom{D-1}{x-1}$,

$$EX = D\sum_x \binom{D-1}{x-1}\binom{N-1-(D-1)}{n-1-(x-1)}\Big/\binom{N}{n} = D\binom{N-1}{n-1}\Big/\binom{N}{n} = \frac{nD}{N}.$$

The middle sum is Vandermonde's identity, obtained by comparing the coefficient of t^{n-1} on both sides of $(1+t)^{D-1}(1+t)^{N-D} = (1+t)^{N-1}$. In the same way $x(x-1)\binom{D}{x} = D(D-1)\binom{D-2}{x-2}$ gives $E[X(X-1)] = n(n-1)\frac{D(D-1)}{N(N-1)}$. Then

$$\operatorname{Var} X = E[X(X-1)] + EX - (EX)^2 = \frac{N-n}{N-1}\,n\frac{D}{N}\Big(1-\frac{D}{N}\Big).$$

■

Example 3.1 (Acceptance sampling)

A lot of $N = 40$ items contains $D = 4$ defectives. We inspect $n = 5$ at random and accept the lot if at most c defectives are found. Then

$$P(X=0) = \binom{36}{5}\Big/\binom{40}{5} = \frac{376992}{658008} \approx 0.573, \qquad P(X=1) = \frac{4\binom{36}{4}}{\binom{40}{5}} = \frac{235620}{658008} \approx 0.358.$$

So $P(X \le 1) \approx 0.931$ and $P(X \le 2) \approx 0.996$. The rule "accept if $X \le 2$" accepts such a lot more than 99% of the time, so to reject a lot with 10% defectives with probability at least 0.99 we would need a smaller acceptance number or a larger sample.

Binomial approximation. If $N \to \infty$ with $D/N = p$ fixed and n fixed, then

$$\binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n} = \binom{n}{x}\frac{D(D-1)\cdots(D-x+1)\cdot(N-D)\cdots(N-D-n+x+1)}{N(N-1)\cdots(N-n+1)}$$
$$\to \binom{n}{x}p^x(1-p)^{n-x}.$$

Drawing without replacement from a huge population is practically the same as drawing with replacement.

3.2 Binomial and multinomial distributions

If each draw is made *with* replacement, the outcomes $Z_1, \dots, Z_n$ are independent with $P(Z_i = 1) = p$: **Bernoulli trials**, $Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$. The number of successes is binomial.

Definition (Binomial)

$X \sim B(n,p)$ if $f(x) = \binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, \dots, n$. Representation: $X \overset{d}{=} Z_1 + \cdots + Z_n$, $Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$.

Proposition 3.2

(a) $EX = np$, $\operatorname{Var} X = np(1-p)$.
(b) $M(t) = (pe^t + q)^n$ with $q = 1 - p$.
(c) If $X_1 \sim B(n_1, p)$ and $X_2 \sim B(n_2, p)$ are independent, then $X_1 + X_2 \sim B(n_1 + n_2, p)$.

Proof. (a) Directly, $EX = \sum x \binom{n}{x} p^x q^{n-x} = np \sum_k \binom{n-1}{k} p^k q^{n-1-k} = np$, and similarly $E[X(X-1)] = n(n-1)p^2$; so $\operatorname{Var} X = n(n-1)p^2 + np - n^2p^2 = np(1-p)$. (Or add the Bernoulli means p and variances $p(1-p)$.) (b) A Bernoulli has mgf $pe^t + q$, and mgfs of independent summands multiply.

(c) The product of the mgfs is $(pe^t + q)^{n_1+n_2}$; uniqueness of mgfs (Theorem 1.7) identifies the distribution. ■

Example 3.2 (Binomial conditioned on its total is hypergeometric)

Let $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $D \le N$. Then the conditional distribution of $X_1 + \cdots + X_D$ given $X_1 + \cdots + X_N = n$ is $H(n; N, D)$. Indeed $X_1 + \cdots + X_D \sim B(D,p)$ and $X_{D+1} + \cdots + X_N \sim B(N-D, p)$ are independent, so

$$P\Big(\sum_{i \le D} X_i = x \,\Big|\, \sum_{i \le N} X_i = n\Big) = \frac{\binom{D}{x} p^x q^{D-x} \binom{N-D}{n-x} p^{n-x} q^{N-D-n+x}}{\binom{N}{n} p^n q^{N-n}} = \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}}.$$

The parameter p cancels.

Definition (Multinomial)

Each of n independent trials falls in category j with probability p_j, $\sum_j p_j = 1$. The counts $\mathbf{X} = (X_1, \dots, X_k)$ satisfy

$$f(\mathbf{x}) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}, \qquad \sum_j x_j = n,$$

written $\mathbf{X} \sim \text{Multi}(n, \mathbf{p})$. Representation: $\mathbf{X} \overset{d}{=} \mathbf{Z}_1 + \cdots + \mathbf{Z}_n$ with $\mathbf{Z}_i \overset{\text{i.i.d.}}{\sim} \text{Multi}(1, \mathbf{p})$ (one-hot vectors). For $k = 2$, $(X, n - X)$ with $X \sim B(n, p_1)$.

Proposition 3.3

(a) $EX_l = np_l$, $\operatorname{Var} X_l = np_l(1-p_l)$, $\operatorname{Cov}(X_l, X_m) = -np_lp_m$ for $l \neq m$; i.e. $\operatorname{Var}\mathbf{X} = n\{\operatorname{diag}(\mathbf{p}) - \mathbf{pp}'\}$.
(b) $M(\mathbf{t}) = (p_1e^{t_1} + \cdots + p_ke^{t_k})^n$.

Proof. For one trial, $Z_{1l} \in \{0,1\}$ with $P(Z_{1l} = 1) = p_l$, and $Z_{1l}Z_{1m} = 0$ for $l \neq m$. So $\operatorname{Cov}(Z_{1l}, Z_{1m}) = -p_lp_m$, and $M_{\mathbf{Z}_1}(\mathbf{t}) = \sum_l p_le^{t_l}$. Sum over n i.i.d. trials (Theorem 2.11). ■

The negative covariances express the constraint $\sum_j X_j = n$.

3.3 Geometric and negative binomial distributions

In Bernoulli trials with success probability p, the trial number W_1 of the first success satisfies $P(W_1 = x) = q^{x-1}p$, $x = 1, 2, \ldots$: the **geometric distribution** Geo(p).

Proposition 3.4

(a) $M_{W_1}(t) = pe^t(1-qe^t)^{-1}$ for $t < -\log q$.
(b) $EW_1 = 1/p$, $\operatorname{Var} W_1 = q/p^2$.
(c) (Memorylessness) $P(W_1 > m+k \mid W_1 > m) = P(W_1 > k)$.

Proof. (a) A geometric series: $\sum_x e^{tx}q^{x-1}p = pe^t\sum_x(qe^t)^{x-1}$.

(b) The cgf is $C(t) = t + \log p - \log(1-qe^t)$. Expanding $-\log(1-qe^t)$ in powers of t gives $C(t) = \frac{1}{p}t + \frac{q}{p^2}\frac{t^2}{2} + \cdots$.

(c) $P(W_1 > k) = q^k$. ■

The trial number W_r of the rth success needs $r-1$ successes in the first $x-1$ trials and a success at trial x:

$$P(W_r = x) = \binom{x-1}{r-1}p^rq^{x-r}, \qquad x = r, r+1, \ldots.$$

This is the **negative binomial distribution** Negbin(r,p). The name comes from the expansion $(1-q)^{-r} = \sum_{k\geq 0}\binom{r+k-1}{k}q^k$; substituting $x = r+k$ shows that the probabilities sum to $p^rp^{-r} = 1$. The waiting times $W_1, W_2 - W_1, \ldots, W_r - W_{r-1}$ are i.i.d. Geo(p), since

$$P(W_1 = x_1, \ldots, W_r - W_{r-1} = x_r) = \prod_{i=1}^{r} q^{x_i-1}p.$$

Hence the representation $\text{Negbin}(r,p) \stackrel{d}{=} \text{Geo}_1(p) \oplus \cdots \oplus \text{Geo}_r(p)$, where $\oplus$ denotes a sum of independent variables.

Proposition 3.5

For $X \sim \text{Negbin}(r,p)$: $EX = r/p$, $\operatorname{Var} X = rq/p^2$, $M(t) = \{pe^t(1-qe^t)^{-1}\}^r$. Independent Negbin($r_i,p$) variables sum to Negbin($\sum r_i, p$).

Proof. Sum r i.i.d. geometrics (Proposition 3.4); multiply mgfs and use uniqueness. ■

3.4 The Poisson distribution

Proposition 3.6 (Poisson approximation of the binomial)

If $n \to \infty$ and $np_n \to \lambda > 0$, then $\binom{n}{x} p_n^x (1-p_n)^{n-x} \to e^{-\lambda}\lambda^x/x!$.

Proof.

$$\binom{n}{x} p_n^x (1-p_n)^{n-x} = \frac{n(n-1)\cdots(n-x+1)}{n^x} \cdot \frac{(np_n)^x}{x!} \cdot \Big(1 - \frac{np_n}{n}\Big)^n (1-p_n)^{-x}.$$

The first factor tends to 1, the second to $\lambda^x/x!$, the third to $e^{-\lambda}$ (since $(1+a_n/n)^n \to e^a$ when $a_n \to a$), and the last to 1. ■

$X \sim \text{Poisson}(\lambda)$ if $f(x) = e^{-\lambda}\lambda^x/x!$, $x = 0, 1, \ldots$.

Example 3.3

A batter hits a home run in 2% of at-bats. Over 300 at-bats, $X \sim B(300, 0.02) \approx \text{Poisson}(6)$. Then $P(X = 10) \approx e^{-6}6^{10}/10! \approx 0.041$; the exact binomial value is 0.040.

Proposition 3.7

(a) $M(t) = \exp\{\lambda(e^t - 1)\}$ for all t.
(b) $EX = \text{Var}\, X = \lambda$.
(c) Independent Poisson(λ_i) variables sum to Poisson($\sum \lambda_i$).

Proof. (a) $\sum_x e^{tx} e^{-\lambda}\lambda^x/x! = e^{-\lambda}e^{\lambda e^t}$.

(b) The cgf $\lambda(e^t - 1) = \lambda \sum_{r \geq 1} t^r/r!$ has every cumulant equal to λ.

(c) Multiply mgfs and use uniqueness. ■

The Poisson process. Let N_t count the events in $[0, t]$, with

(a) (stationarity) $N_{s+t} - N_s \overset{d}{=} N_t$ and $N_0 = 0$;
(b) (independent increments) counts on disjoint intervals are independent;
(c) (proportionality) $P(N_h = 1) = \lambda h + o(h)$;
(d) (rareness) $P(N_h \geq 2) = o(h)$ as $h \downarrow 0$.

Theorem 3.8

Under (a)–(d), $N_t \sim \text{Poisson}(\lambda t)$.

Proof. Let $g(x, t) = P(N_t = x)$. Splitting according to the increment on $(t, t+h]$ and using (a), (b), (d),

$$g(x, t+h) = g(x, t)P(N_h = 0) + g(x-1, t)P(N_h = 1) + o(h) = g(x, t)(1 - \lambda h) + g(x-1, t)\lambda h + o(h).$$

Hence $\partial_t g(x, t) = -\lambda g(x, t) + \lambda g(x-1, t)$, i.e. $\frac{d}{dt}\{e^{\lambda t} g(x, t)\} = \lambda e^{\lambda t} g(x-1, t)$, with $g(0, 0) = 1$ and $g(x, 0) = 0$ for $x \geq 1$. For $x = 0$ this gives $g(0, t) = e^{-\lambda t}$. Inductively, if $g(x-1, t) = e^{-\lambda t}(\lambda t)^{x-1}/(x-1)!$, then $e^{\lambda t} g(x, t) = \int_0^t \lambda(\lambda s)^{x-1}/(x-1)!\, ds = (\lambda t)^x/x!$. ■

Heuristically: split $[0, t]$ into n pieces; each contains an event with probability about $\lambda t/n$, independently, so $N_t \approx B(n, \lambda t/n) \to \text{Poisson}(\lambda t)$.

Example 3.4

A cable has on average 0.04 flaws per meter. In 150 meters the number of flaws is Poisson(6), so $P(\text{at least } 10) = 1 - \sum_{x\le 9} e^{-6}6^x/x! \approx 0.084$.

3.5 Exponential and gamma distributions

In a Poisson process with rate λ, the waiting time W_1 for the first event satisfies $\{W_1 > t\} = \{N_t = 0\}$. So $P(W_1 \le t) = 1 - e^{-\lambda t}$, and W_1 has density $\lambda e^{-\lambda t}$, $t > 0$: the **exponential distribution** $\text{Exp}(1/\lambda)$, with mean $1/\lambda$.

Proposition 3.9

$M_{W_1}(t) = (1 - t/\lambda)^{-1}$ for $t < \lambda$; $EW_1 = 1/\lambda$, $\text{Var}\, W_1 = 1/\lambda^2$; and $P(W_1 > s + t \mid W_1 > s) = P(W_1 > t)$.

Proof. $\int_0^\infty e^{tx}\lambda e^{-\lambda x}\,dx = \lambda/(\lambda - t)$. The cgf $-\log(1 - t/\lambda) = t/\lambda + \frac{1}{2}(t/\lambda)^2 + \cdots$ gives the mean and variance. Memorylessness: $e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. ■

For the rth event, $\{W_r > t\} = \{N_t \le r-1\}$, so $P(W_r \le t) = 1 - \sum_{k=0}^{r-1} e^{-\lambda t}(\lambda t)^k/k!$. Differentiating, the sum telescopes:

$$f_{W_r}(t) = \lambda e^{-\lambda t}\Big\{\sum_{k=0}^{r-1}\frac{(\lambda t)^k}{k!} - \sum_{k=1}^{r-1}\frac{(\lambda t)^{k-1}}{(k-1)!}\Big\} = \frac{\lambda^r t^{r-1}e^{-\lambda t}}{(r-1)!}, \qquad t > 0.$$

Allowing any shape $\alpha > 0$ gives the gamma family.

Definition (Gamma)

$X \sim \text{Gamma}(\alpha, \beta)$ if $f(x) = \dfrac{1}{\Gamma(\alpha)\beta^\alpha}x^{\alpha-1}e^{-x/\beta}$, $x > 0$, where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}\,dx$. Here α is the shape and β the scale, and $\text{Exp}(\beta) = \text{Gamma}(1, \beta)$.

Integration by parts gives $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$, so $\Gamma(n) = (n-1)!$. Also $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (substitute $x = z^2/2$ and use $\int \phi = 1$).

Proposition 3.10

(a) $EX = \alpha\beta$, $\text{Var}\, X = \alpha\beta^2$.
(b) $M(t) = (1 - \beta t)^{-\alpha}$ for $t < 1/\beta$.
(c) Independent $\text{Gamma}(\alpha_i, \beta)$ variables with a common scale sum to $\text{Gamma}(\sum \alpha_i, \beta)$. In particular $\text{Gamma}(r, \beta) \overset{d}{=} Z_1 + \cdots + Z_r$ with $Z_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\beta)$.

Proof. (a) Substituting $y = x/\beta$, $EX^k = \beta^k\Gamma(\alpha + k)/\Gamma(\alpha)$, which gives $\alpha\beta$ and $\alpha(\alpha + 1)\beta^2$.

(b) Substitute $y = (1/\beta - t)x$ in $\int e^{tx}f(x)\,dx$.

(c) Multiply the mgfs. ■

In a Poisson process the inter-event times $W_1, W_2 - W_1, \ldots$ are i.i.d. $\text{Exp}(1/\lambda)$, and W_r is their sum.

3.6 The normal distribution

Definition (Normal)

$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal density, with cdf Φ. $X \sim N(\mu, \sigma^2)$ has density $\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$.

$\int \phi = 1$: $(\int e^{-z^2/2}\,dz)^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi$.

Proposition 3.11

(a) $EX = \mu$, $\operatorname{Var} X = \sigma^2$.
(b) $M(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$ for all t.
(c) If $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof. (a) With $z = (x-\mu)/\sigma$: $EX = \mu + \sigma \int z\phi(z)\,dz = \mu$ by symmetry. Also $\int z^2\phi = 2\int_0^\infty z^2\phi(z)\,dz = \frac{2}{\sqrt{\pi}}\Gamma(\frac{3}{2}) = 1$ (substitute $y = z^2/2$), using $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi}$.

(b) $\int e^{tz}\phi(z)\,dz = e^{t^2/2}\int \phi(z-t)\,dz = e^{t^2/2}$ (complete the square). Then $Ee^{t(\sigma Z+\mu)} = e^{\mu t + \sigma^2 t^2/2}$.

(c) Multiply the mgfs. ■

Proposition 3.12 (Representation)

(a) $X \sim N(\mu, \sigma^2) \Rightarrow aX + b \sim N(a\mu + b, a^2\sigma^2)$.
(b) $X \sim N(\mu, \sigma^2) \iff (X-\mu)/\sigma \sim N(0,1) \iff X \overset{d}{=} \sigma Z + \mu$ with $Z \sim N(0,1)$.

Proof. $M_{aX+b}(t) = e^{bt} M_X(at) = \exp\{(a\mu + b)t + \frac{1}{2}a^2\sigma^2 t^2\}$;

(b) is the special case $a = 1/\sigma$, $b = -\mu/\sigma$. ■

So $P(X \le x) = \Phi((x-\mu)/\sigma)$, and $\Phi(-z) = 1 - \Phi(z)$ by symmetry. The upper α point z_α ($P(Z > z_\alpha) = \alpha$) gives the upper α point $\mu + \sigma z_\alpha$ of $N(\mu, \sigma^2)$; e.g. $z_{0.05} = 1.645$, $z_{0.025} = 1.96$.

Example 3.5

$X \sim N(10, 9)$. Then $P(7 < X \le 16) = \Phi(2) - \Phi(-1) = 0.9772 - 0.1587 = 0.8185$ and $P(X > 13) = 1 - \Phi(1) = 0.1587$. The 95th percentile is $10 + 3(1.645) = 14.94$ and the 2.5th percentile is $10 - 3(1.96) = 4.12$.

The normal density arises as the limit of standardized binomial probabilities (de Moivre–Laplace):

$$\sum_{a \le (x-np)/\sqrt{np(1-p)} \le b} \binom{n}{x} p^x (1-p)^{n-x} \to \int_a^b \phi(z)\,dz.$$

This is a special case of the central limit theorem (Chapter 5).

Summary

distribution	pmf/pdf	representation
$B(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	sum of n i.i.d. Bernoulli(p)
Multi($n,\mathbf{p}$)	$\frac{n!}{\prod x_j!}\prod p_j^{x_j}$	sum of n i.i.d. Multi($1,\mathbf{p}$)
Negbin(r,p)	$\binom{x-1}{r-1}p^r(1-p)^{x-r}$	sum of r i.i.d. Geo(p)
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	limit of $B(n,\lambda/n)$
Gamma(r,β)	$\frac{x^{r-1}e^{-x/\beta}}{\Gamma(r)\beta^r}$	sum of r i.i.d. Exp(β)
$N(\mu,\sigma^2)$	$\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$	$\mu+\sigma Z$, $Z\sim N(0,1)$